

Integrated Flood Hazard and Settlement Vulnerability Assessment in Khagaria District, Bihar Using Geospatial Techniques and the AHP Method

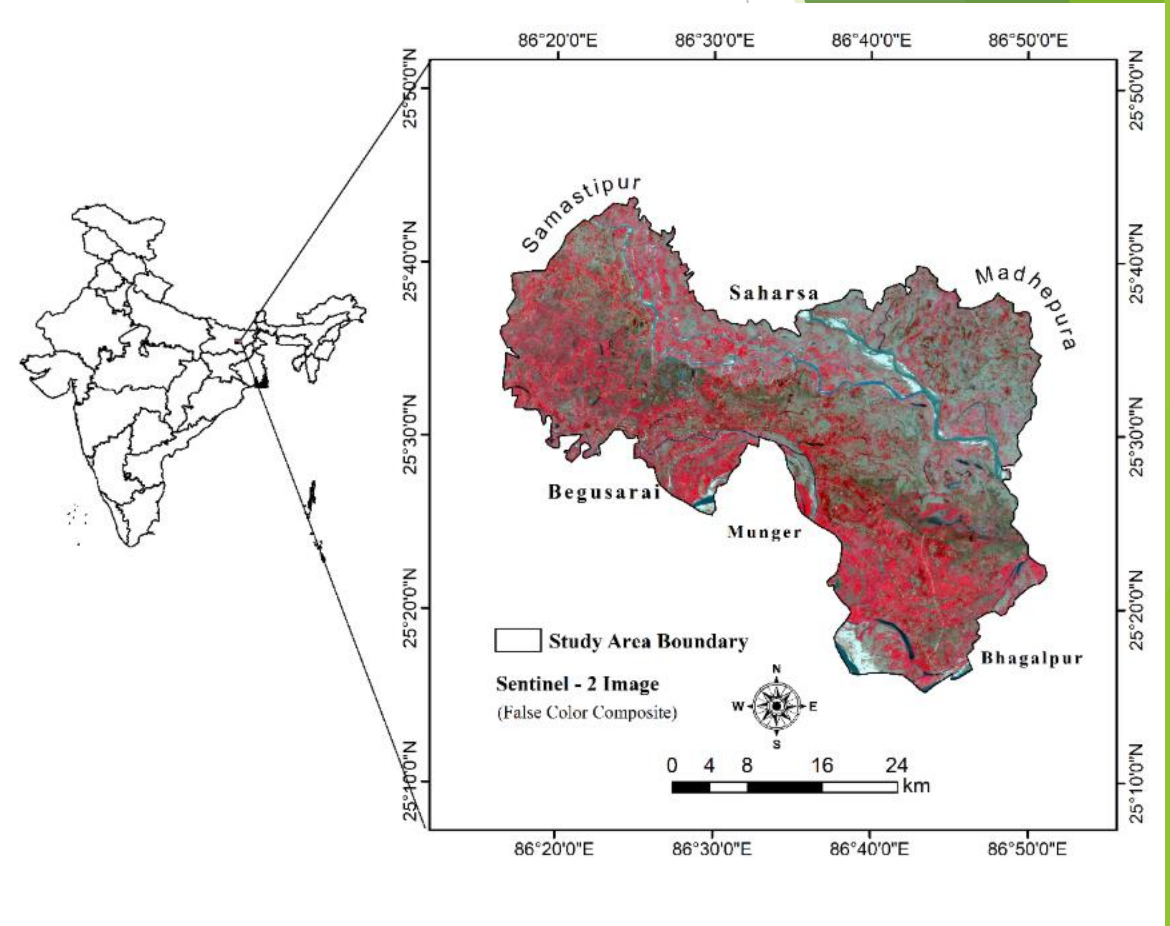
Introduction

The flood history of Khagaria district in Bihar has been catastrophic, with over 74% of the region susceptible to flooding. The significant floods that transpired in 1954, 1957, 1974, 1987, 2004, 2007, 2017, 2020, 2022, and 2024 profoundly affected individuals' lives, properties, and the district's agricultural sector. The current research seeks to employ a multicriteria decision-making framework within a Geographical Information System (GIS), integrating remotely sensed data, to delineate flood-prone areas in Khagaria district. The research employed an 11x11 decision matrix to assess the significance of each parameter in generating the final flood map via the Analytic Hierarchy Process (AHP). The parameters included are elevation, slope, rainfall, drainage density (DD), distance from river (DFR), topographic wetness index (TWI), stream power index (SPI), land use/land cover (LULC), normalised difference vegetation index (NDVI), lithology, and ruggedness index.

Aims, Objective, Study Area and Dataset

Aim and objective - The present study aims to identify flood-prone areas in the Khagaria district and propose appropriate flood mitigation strategies by employing a multicriteria decision-making approach inside a GIS framework, utilising remotely sensed data.

SNO.	DATA TYPES	SOURCES
1.	Rainfall Data	Climate Research Unit (CRU) https://crudata.uea.ac.uk/cru/data/hrg/ .
2.	SRTM-DEM (30m resolution)	USGS Earth Explorer website (https://earthexplorer.usgs.gov/).
3.	Sentinal-1 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
4.	Sentinal-2 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
6.	Lithology Data	Geological Survey of India (GSI), https://www.gsi.gov.in/webcenter/portal/OCBIS



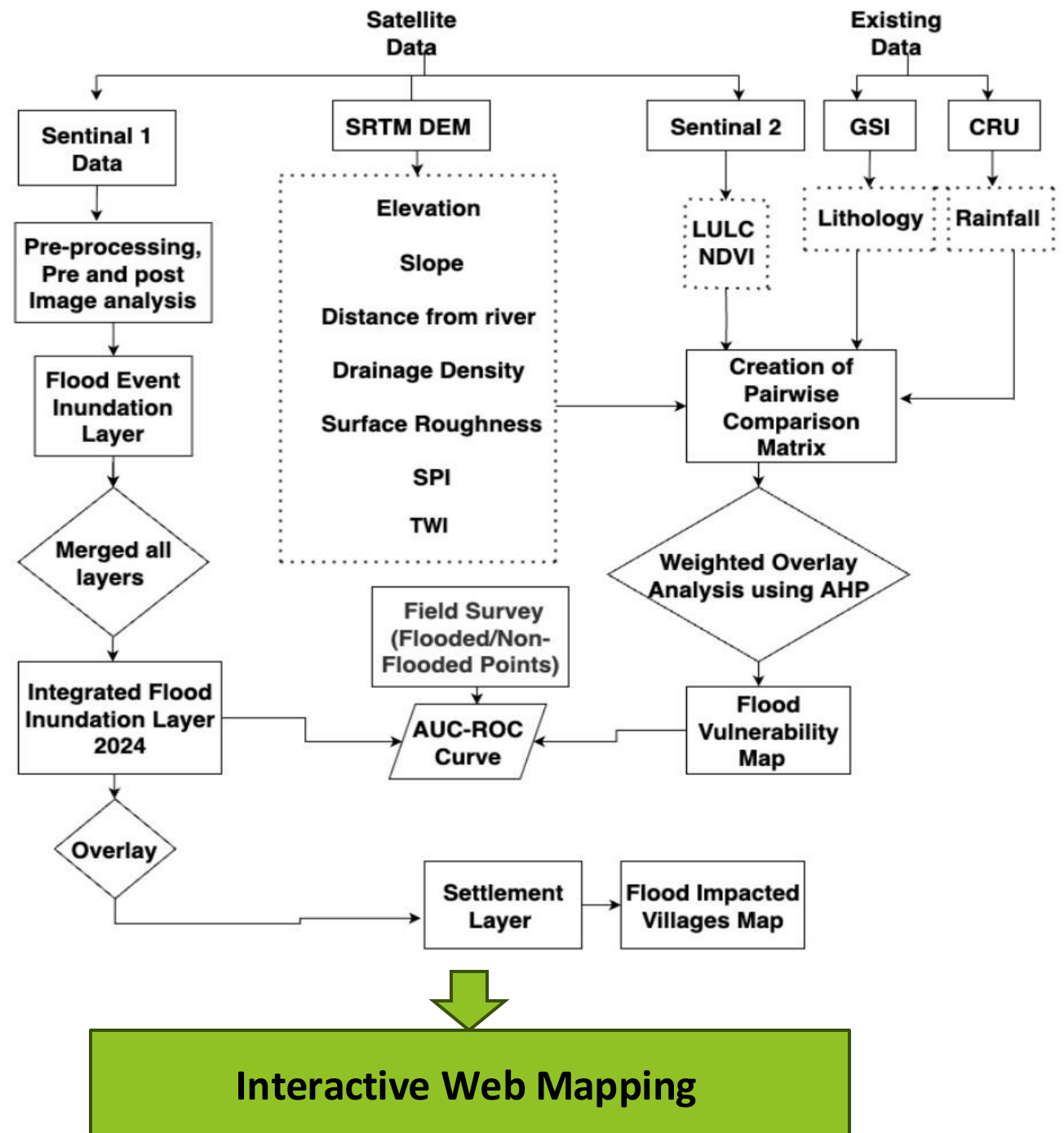
Study Area

Methodology

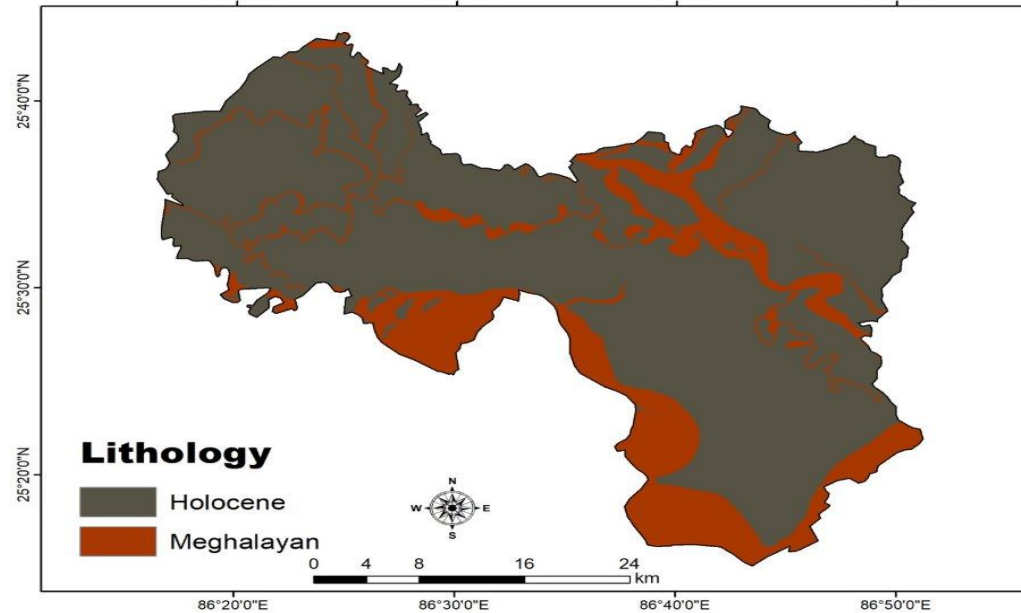
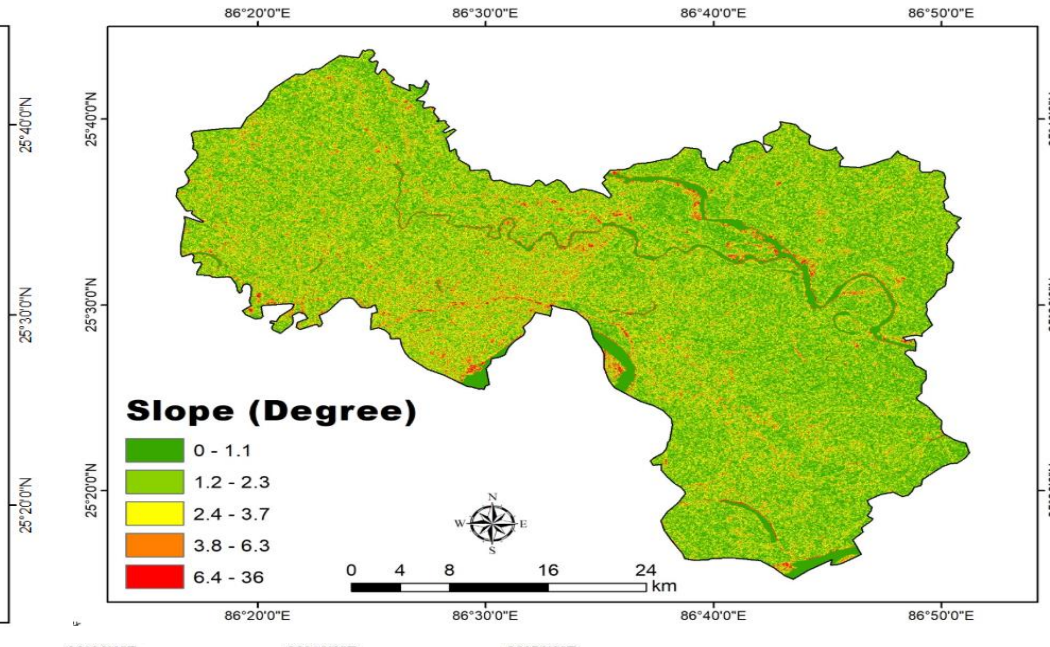
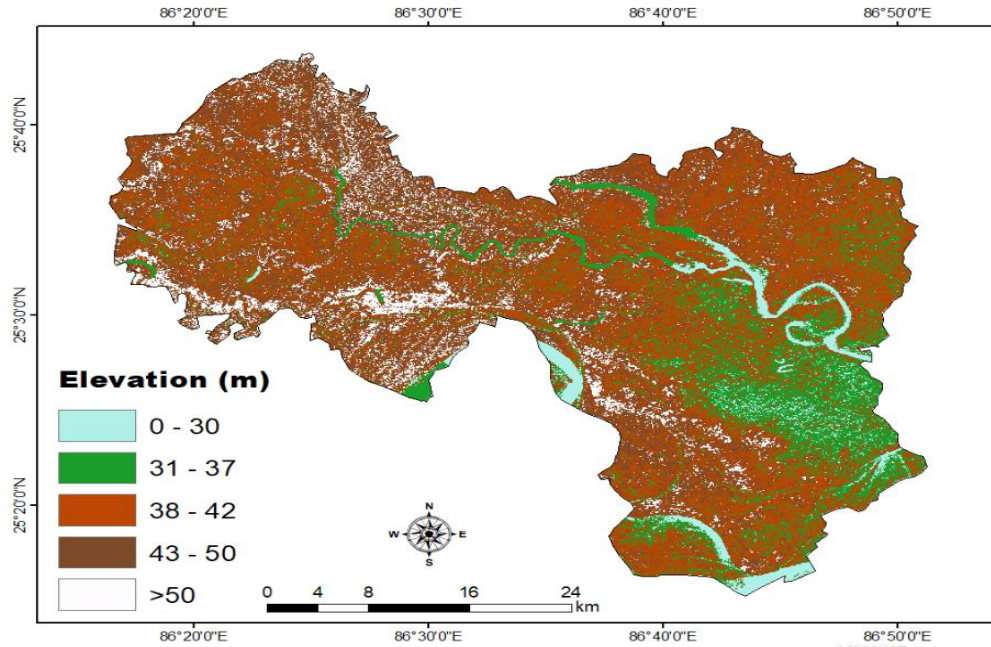
This workflow shows how flood vulnerability is mapped using satellite and existing data. Sentinel and DEM data are processed to derive factors like elevation, slope, drainage density, and land use, while rainfall and lithology are also included.

These factors are weighted using AHP (pairwise comparison) and combined through overlay analysis to produce a flood vulnerability map.

After getting the flood map, an interactive flood GIS dashboard is made using the files- Index.html, style.css, script.js.

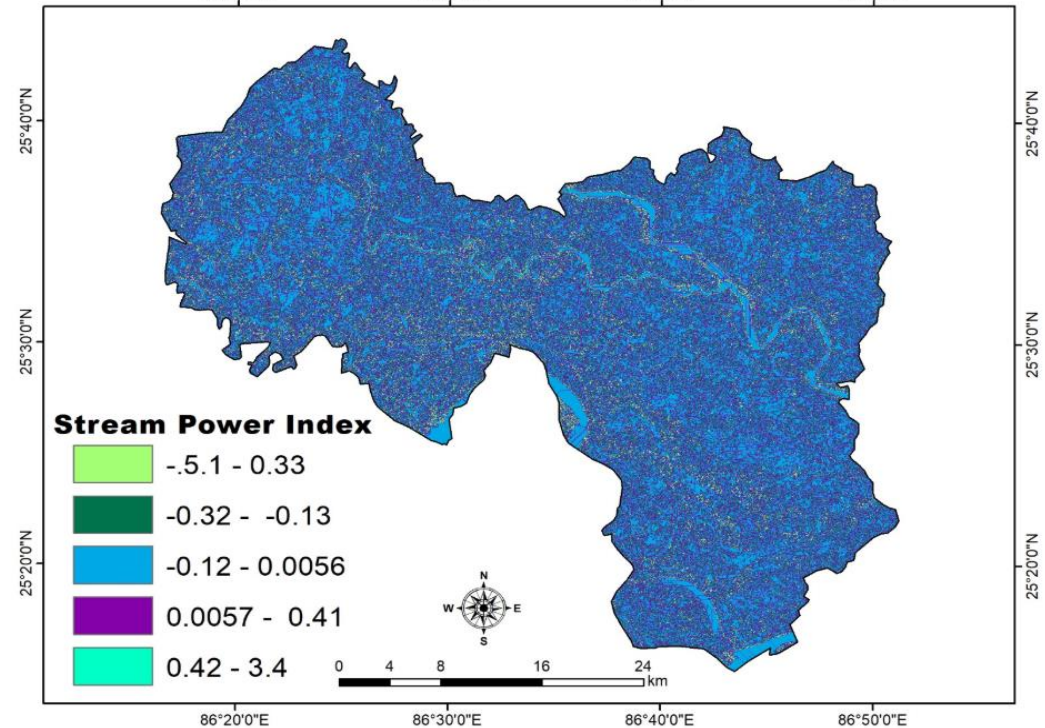
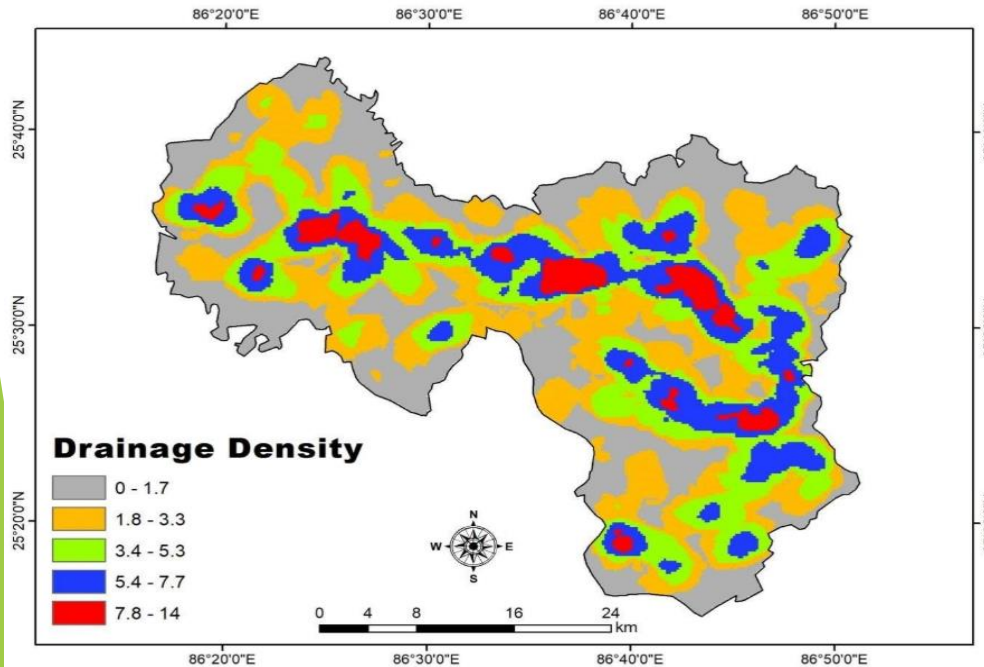
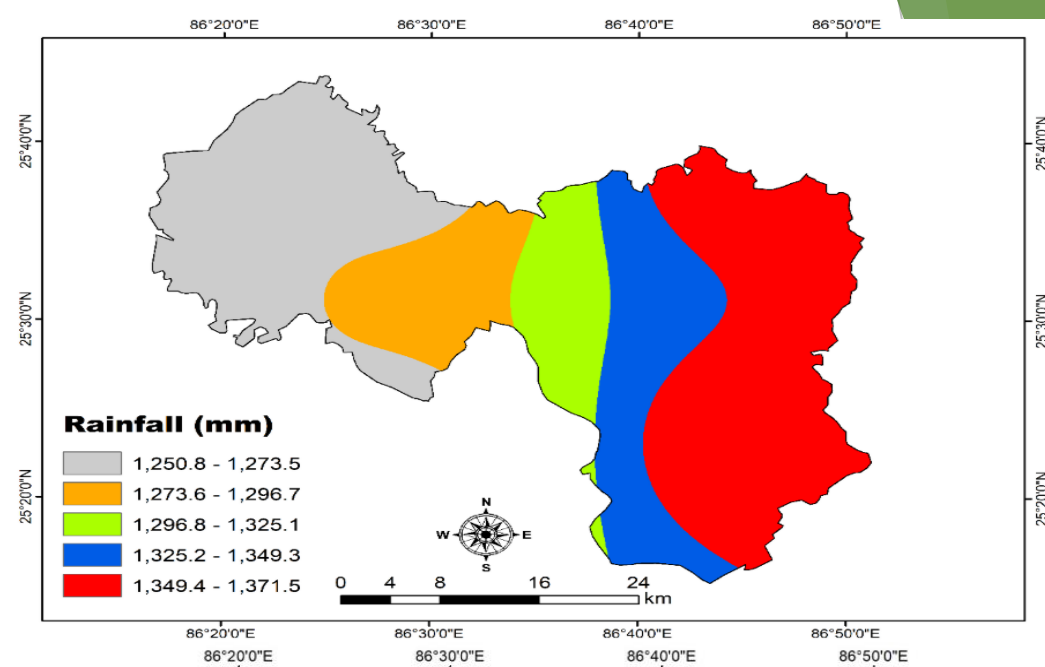
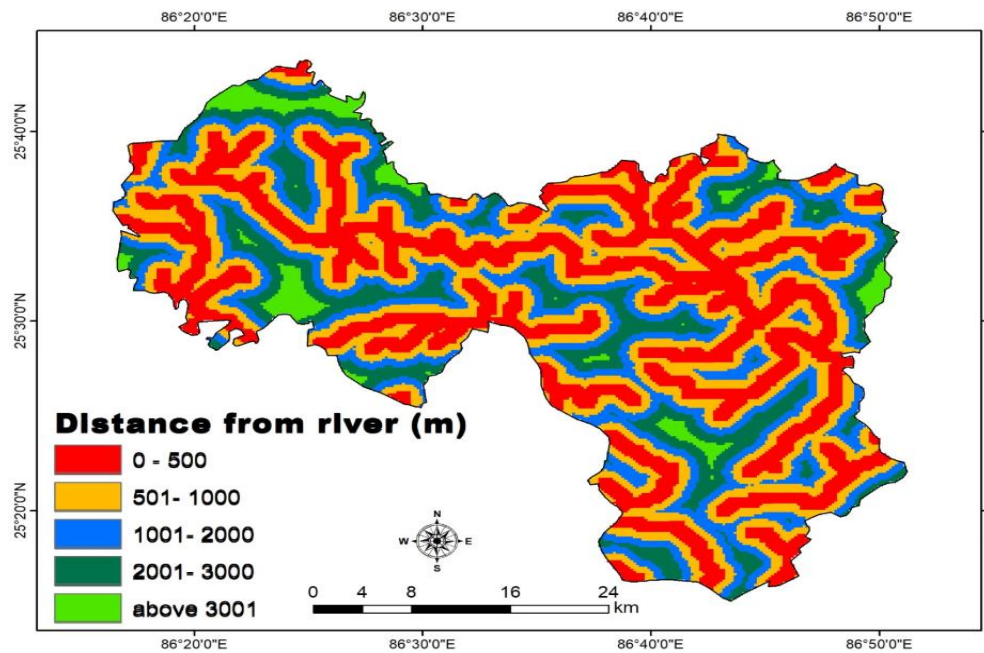


Parameters Used for Flood Vulnerability Assessment (AHP)



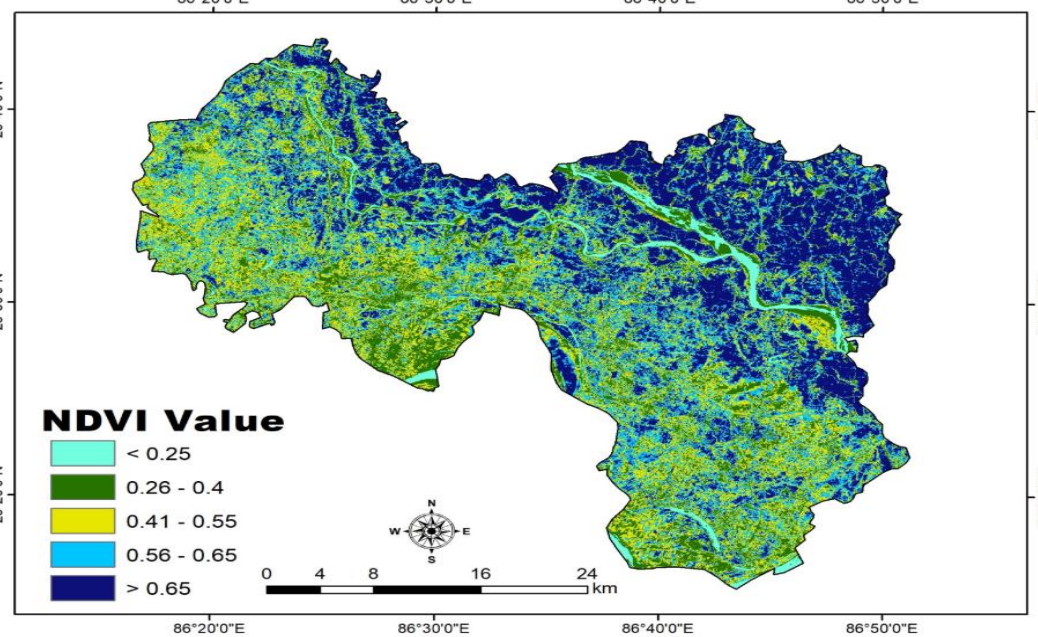
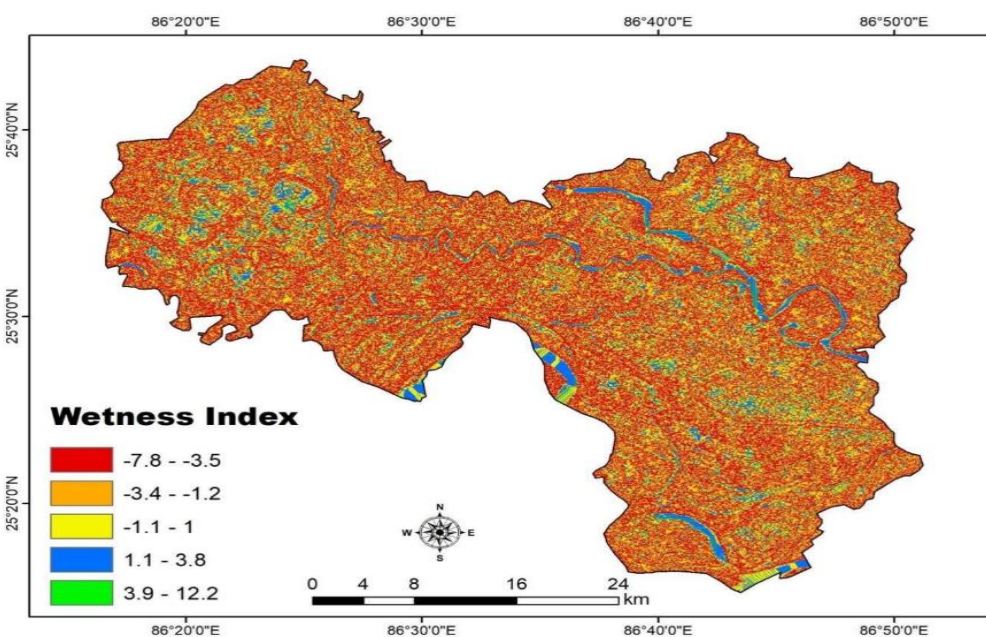
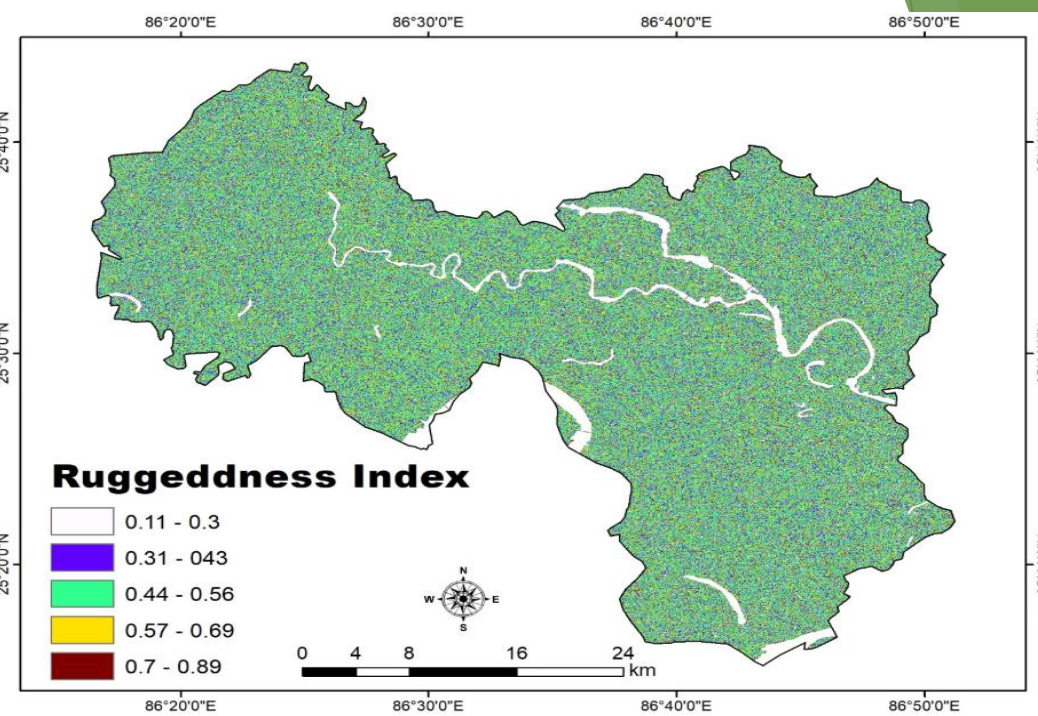
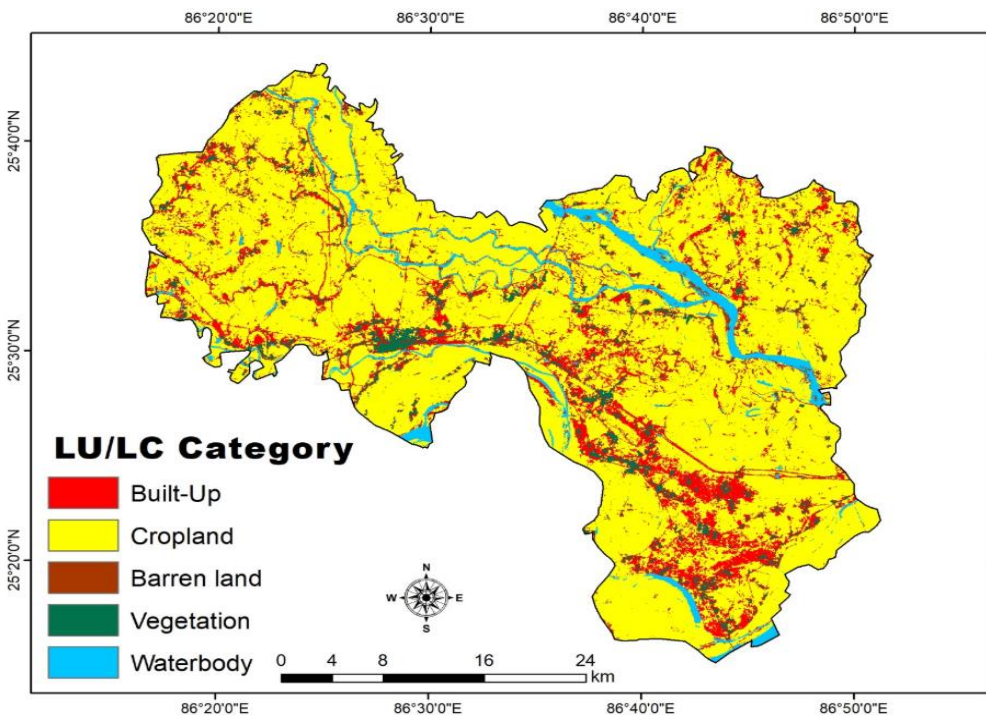
Terrain Parameters:

- Elevation
- Slope
- Lithology



Hydrological Parameters: Rainfall

- Distance from River
- Drainage Density
- Stream Power Index



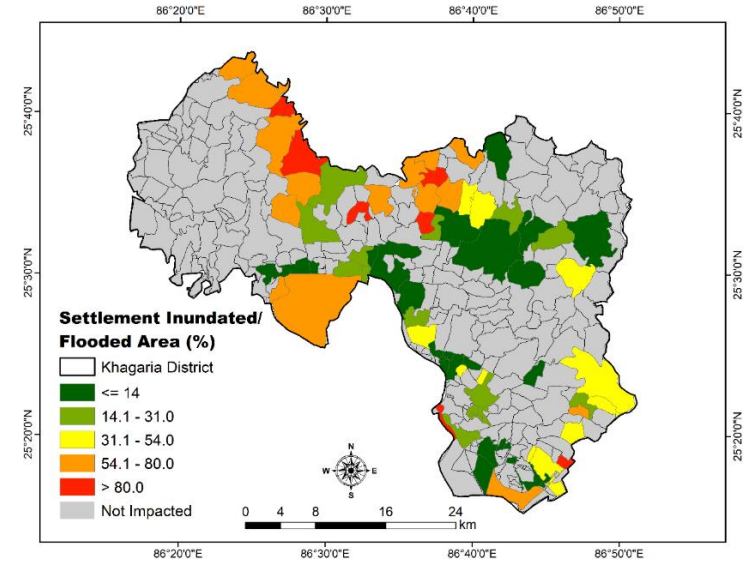
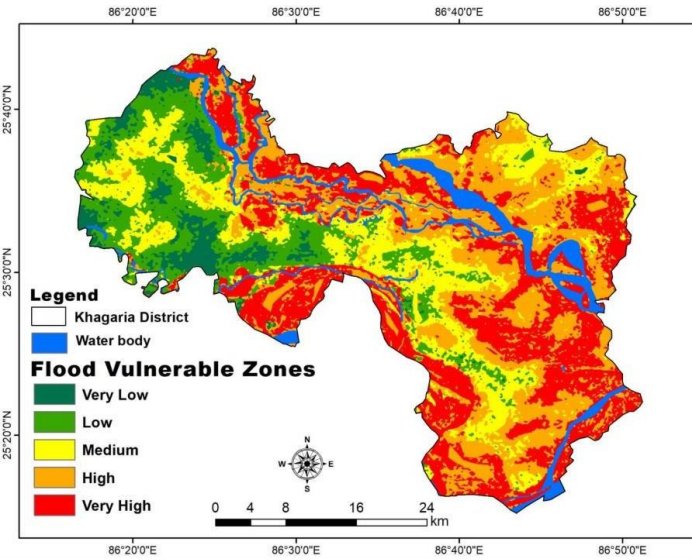
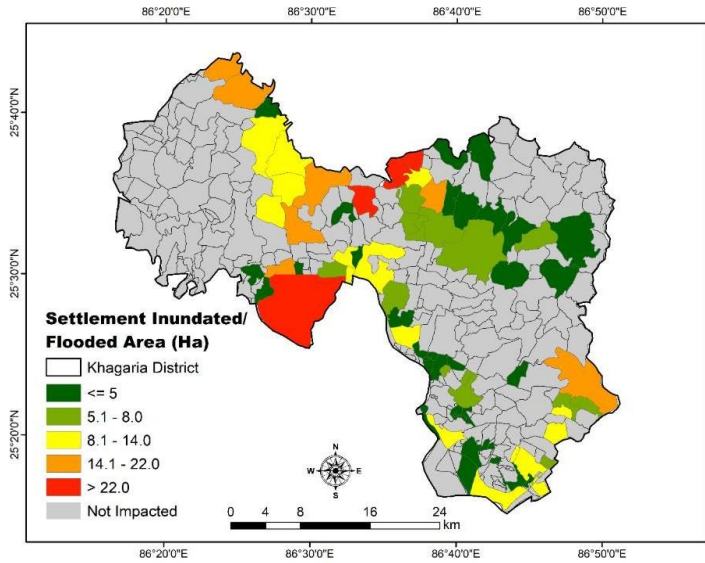
Environmental Parameters:

- NDVI
- LULC

Terrain Parameters:

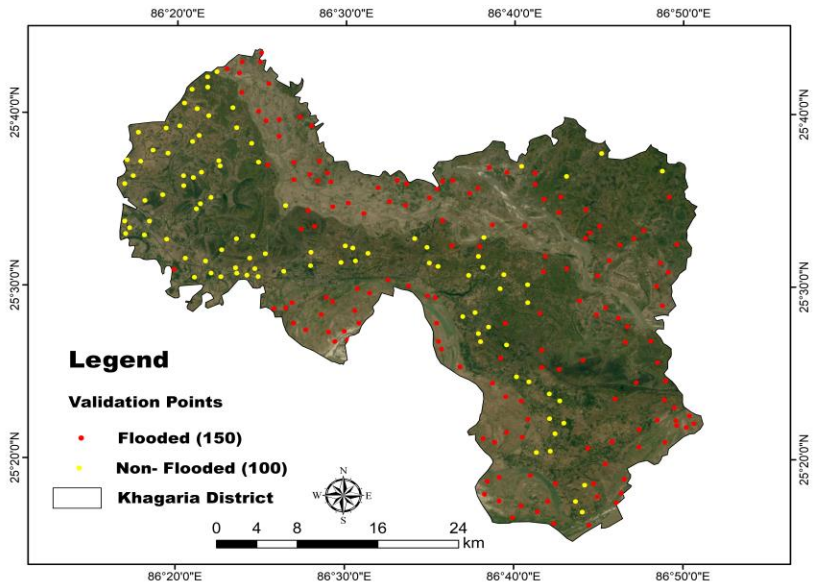
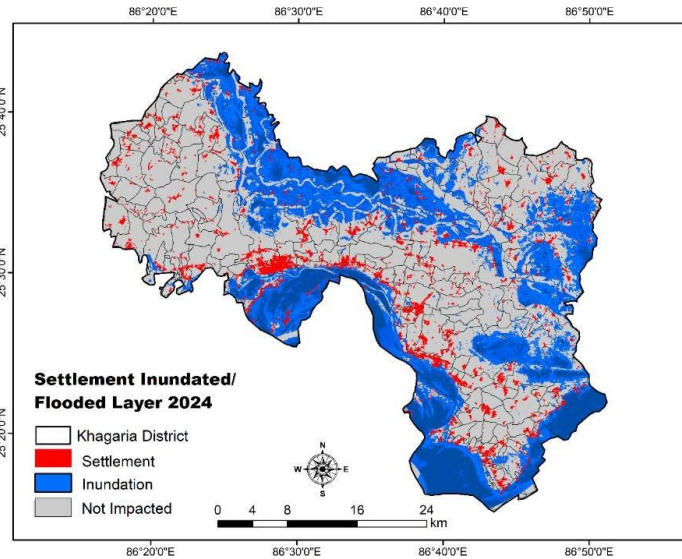
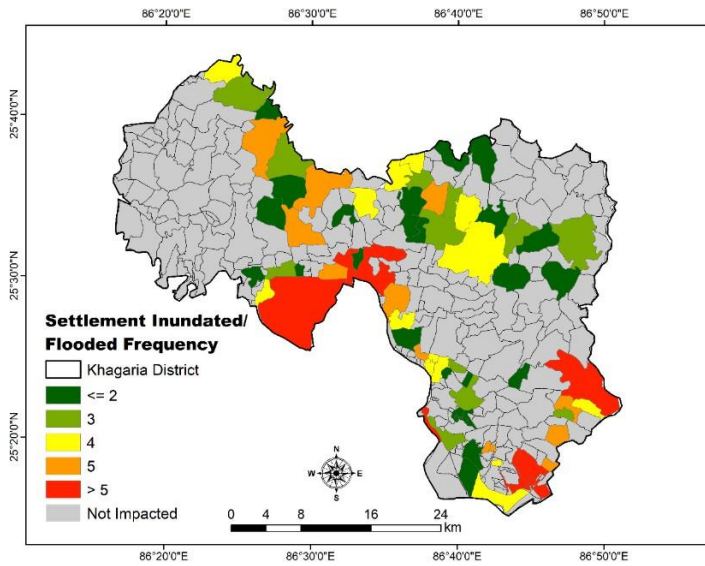
- Topographic Wetness Index
- Ruggedness Index

Flood Vulnerability and Settlement Vulnerability Maps



Where green colour indicates low vulnerables zones, Yellow with Moderate, Orange with high and red with very high Vulnerable zones.

Settlement Vulnerability and Flood Validation Maps

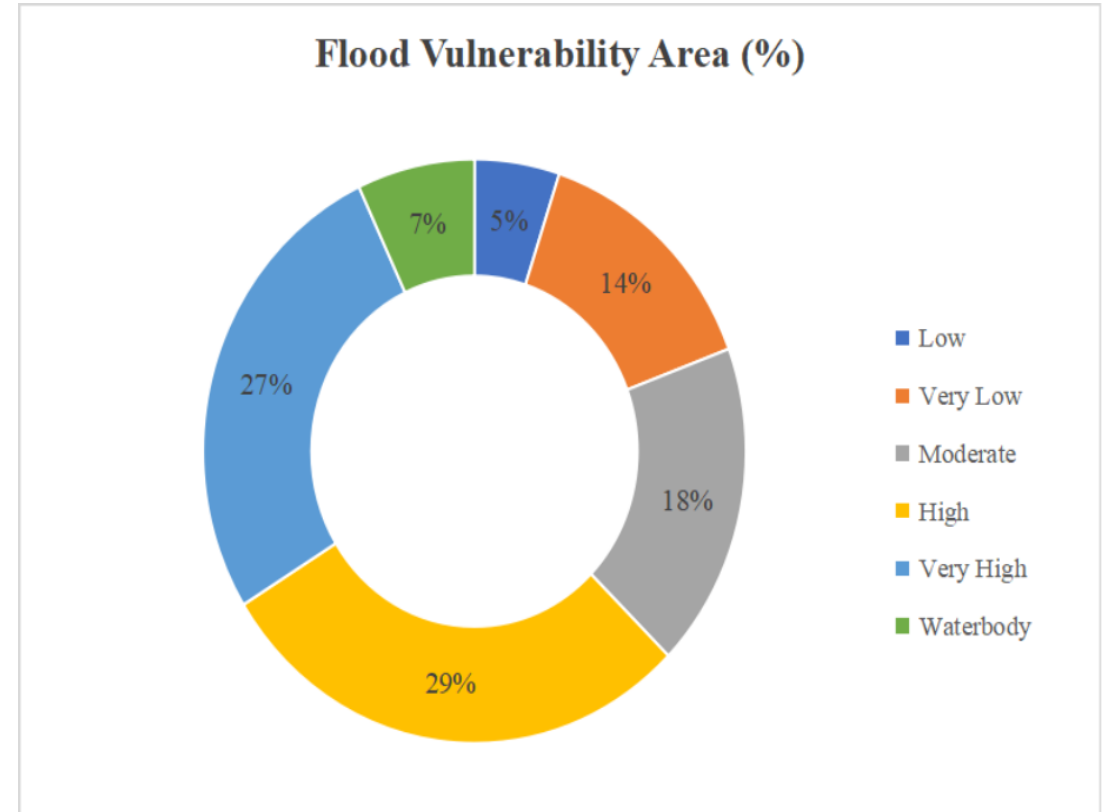


Red Colour shows settlement while blue colour shows Flood Inundation in Settlement inundation map

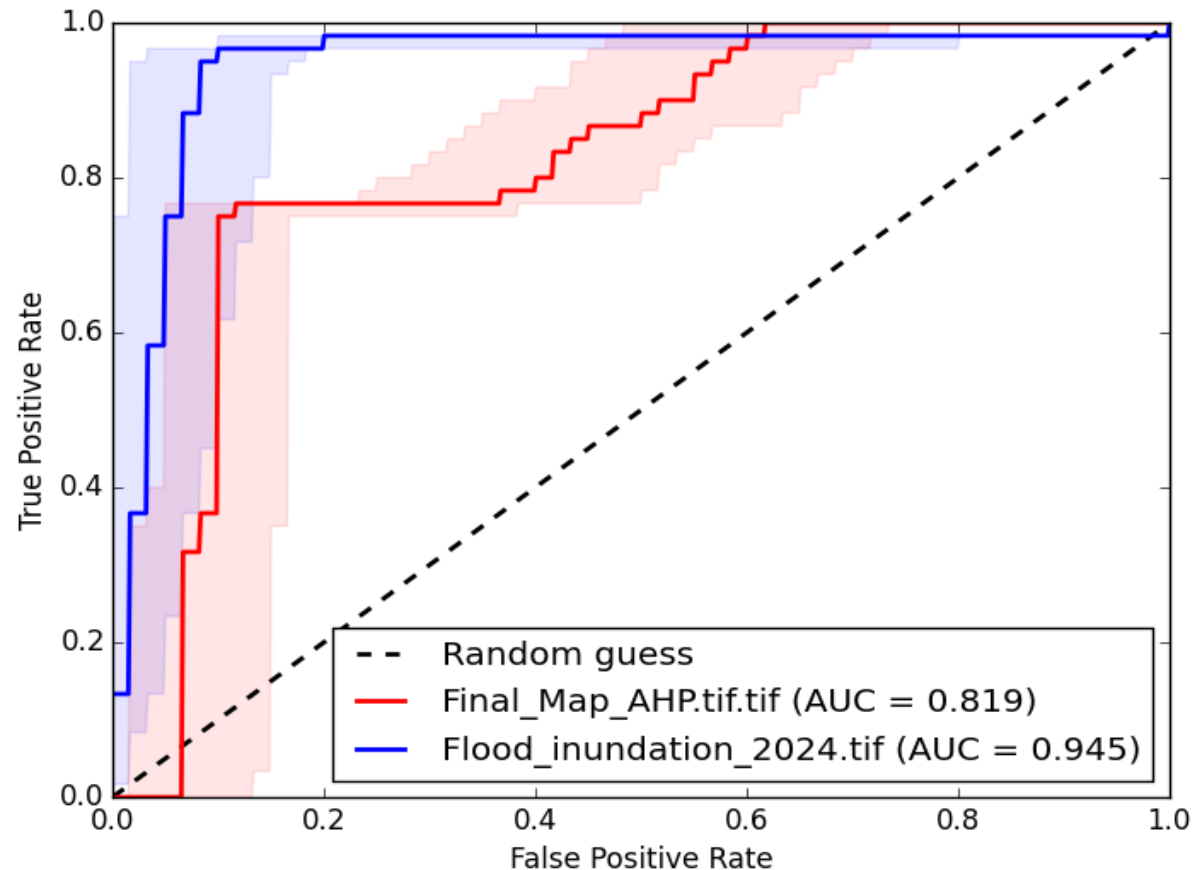
Yellow points indicate non-flooded points and Red indicates Flooded points in Flood Validation Map-250 points.

Flood vulnerable areas with their percentage

Sr. No	Class	Area (Sq.km)
	Very Low	75.362
	Low	212.323
	Moderate	268.844
	High	429.909
	Very High	399.502
	Waterbody	104.221
	Total	1490.16



Model Validation using ROC Curve



ROC and AUC curve

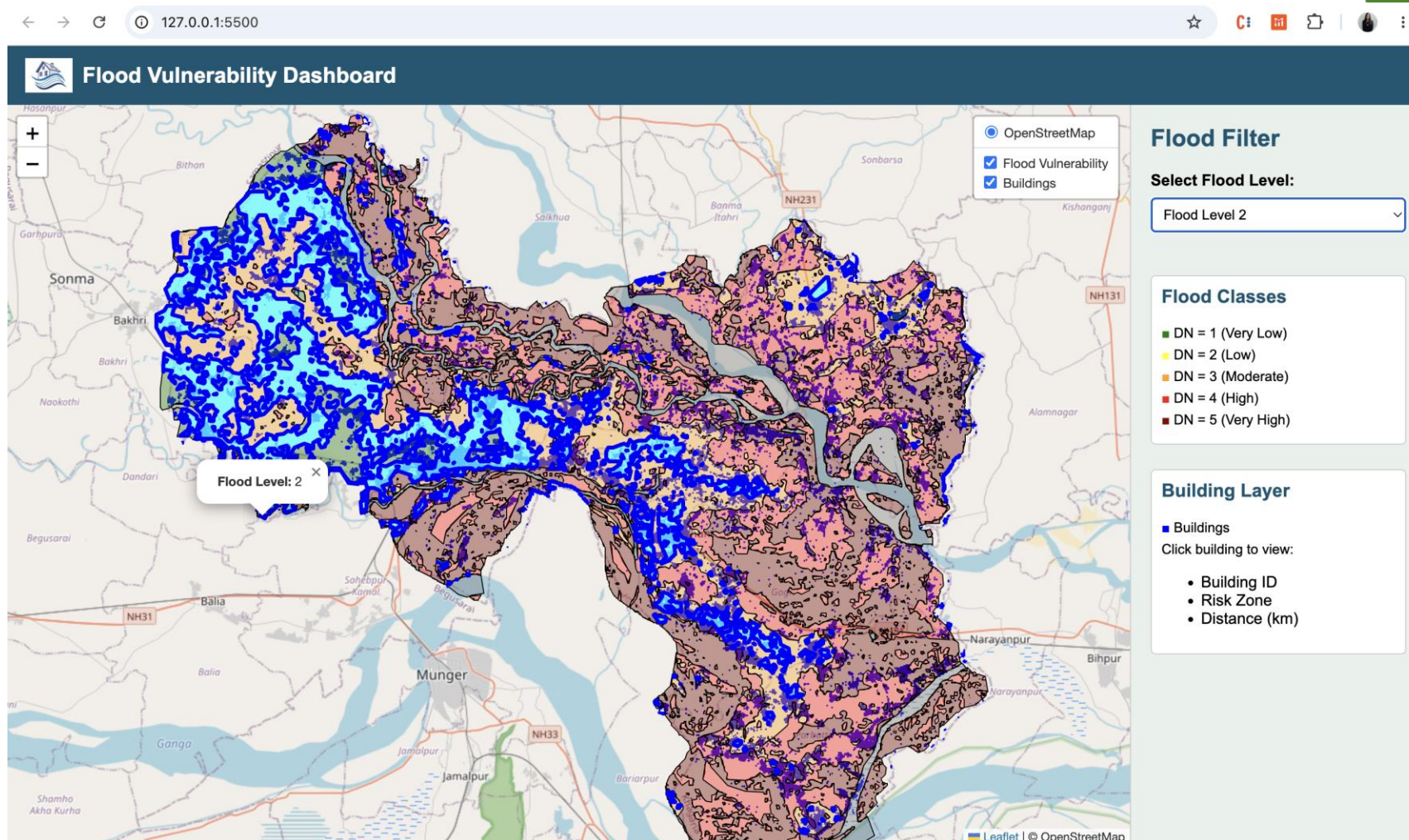
- The red curve represents the model performance (FSM_AHP.tif), while the dashed diagonal line shows random guessing (no predictive power).
- The curve rises steeply toward the top-left corner, indicating high True Positive Rate (sensitivity) with low False Positive Rate.
- The AUC (Area Under Curve) = 0.789, which indicates acceptable and reliable prediction accuracy in distinguishing between flood-prone and non-flood-prone areas.
- Since the curve is far above the diagonal line, the model performs significantly better than random classification and is highly reliable for prediction.

Script for
joining Flood
layer to
building layer
for accessing
the risk zones
for flood

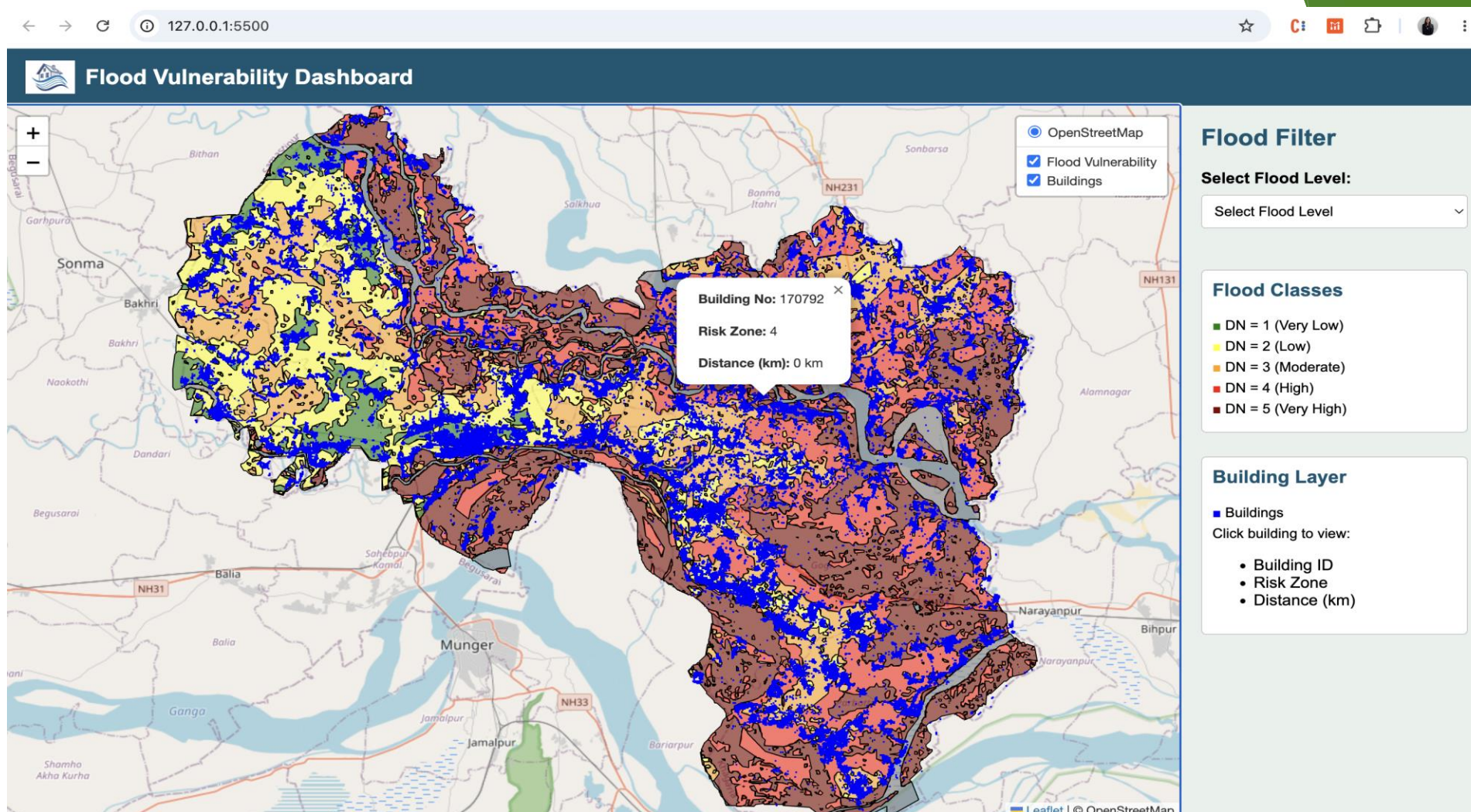
► <https://github.com/madhubalapriya2-debug/Spatialjoin->

	latitude	longitude	area_in_me	confidence	full_plus_	DN	risk_zone	distance_m	distance_k	
9	25.28621249...	86.73254740...	18.83940000...	0.6518	7MQ87PPM+...	3.00000000...	3.00000000...	0	0	
10	25.55104732...	86.29973128...	17.34290000...	0.6748	7MQ8H72X+...	1.00000000...	1.00000000...	0	0	
11	25.5944543...	86.6049426...	13.12580000...	0.6721	7MQ8HJV3+...	4.00000000...	4.00000000...	0	0	
12	25.56663198...	86.70590187...	37.81830000...	0.8185	7MQ8HP84+...	4.00000000...	4.00000000...	0	0	
13	25.3245008...	86.6834247...	81.27559999...	0.8158	7MQ88MFM+...	4.00000000...	4.00000000...	0	0	
14	25.62259166...	86.3032082...	10.29820000...	0.7244	7MQ8J8F3+2...	2.00000000...	2.00000000...	0	0	
15	25.6023956...	86.79478353...	53.2832000...	0.7518	7MQ8JQ2V+...	1.00000000...	1.00000000...	0	0	
16	25.6023956...	86.79478353...	53.2832000...	0.7518	7MQ8JQ2V+...	3.00000000...	3.00000000...	0	0	
17	25.5039543...	86.47297643...	101.6141999...	0.7551	7MQ8GF3F+...	2.00000000...	2.00000000...	0	0	
18	25.44705902...	86.6494656...	30.4287999...	0.7318	7MQ8CJWX+...	3.00000000...	3.00000000...	0	0	
19	25.3268953...	86.75579310...	18.33700000...	0.7512	7MQ88QG4+...	4.00000000...	4.00000000...	0	0	
20	25.59760203...	86.78677059...	80.6835999...	0.7251	7MQ8HQXP+...	3.00000000...	3.00000000...	0	0	
21	25.5423823...	86.34189372...	19.78559999...	0.7803	7MQ8G8RR+...	3.00000000...	3.00000000...	0	0	
22	25.48934148...	86.61554798...	16.18649999...	0.7027	7MQ8FJQ8+...	4.00000000...	4.00000000...	0	0	
23	25.37941895...	86.73654123...	19.61750000...	0.7444	7MQ89PHP+...	4.00000000...	4.00000000...	0	0	
24	25.64931486...	86.35883715...	37.84680000...	0.7488	7MQ8J9X5+...	3.00000000...	3.00000000...	0	0	
25	25.42911412...	86.64392961...	51.23160000...	0.7877	7MQ8CJHV+...	4.00000000...	4.00000000...	0	0	
26	25.37858321...	86.7909239...	26.5765999...	0.6713	7MQ89QHR+...	4.00000000...	4.00000000...	0	0	
27	25.56915929...	86.32414172...	18.80819999...	0.6921	7MQ8H89F+...	3.00000000...	3.00000000...	0	0	
28	25.60549981...	86.51814373...	12.18510000...	0.6732	7MQ8JG49+...	5.00000000...	5.00000000...	0	0	
29	25.37558817...	86.77300694...	7.887500000...	0.6875	7MQ89QGF+...	4.00000000...	4.00000000...	0	0	
30	25.48607573...	86.5233472...	33.3680000...	0.7632	7MQ8FGPF+...	4.00000000...	4.00000000...	0	0	
31	25.51150657...	86.47791205...	126.4586999...	0.8637	7MQ8GF6H+...	2.00000000...	2.00000000...	0	0	
32	25.32640874...	86.68555179...	77.81380000...	0.8246	7MQ88MGP+...	4.00000000...	4.00000000...	0	0	
33	25.46027670...	86.61099704...	64.66100000...	0.6529	7MQ8FJ66+4...	4.00000000...	4.00000000...	0	0	
34	25.31424593...	86.71507762...	60.4898000...	0.8809	7MQ88P78+...	4.00000000...	4.00000000...	0	0	
35	25.40858018...	86.67179654...	49.7288000...	0.8392	7MQ8CM5C+...	3.00000000...	3.00000000...	0	0	
	25.40765077...	86.66666105...	50.00010000...	0.7054	7MQ8CM5C+...	1.00000000...	1.00000000...	0	0	

The attribute table shows building shapefile for Khagaria containing building number and which building falls into which flood risk zone from very high to very low (1-5).



WEB Mapping - Risk Zone Highlight and Pop ups



**WEB Mapping - Building NO., Risk Zone,
Distance from the nearest Flood Zone**

Scripts File for WEB GIS Flood Dashboard

<https://github.com/madhubalapriya2-debug/flood-web-mapping-scripts>

Conclusion

- ▶ **Key Vulnerable Villages**
- ▶ Villages identified as extremely flood-vulnerable include: Imadpur, Dariapur, Siswa, Siadatpur Aguan English, Siromanharapur, Shirinia, Duraja, Marachi, Katghara, Chandwa, Ragdhan Karari, Rampur, Durgapur, and Sathman.
- ▶ These villages fall in very high flood vulnerability zones due to their geographical and environmental conditions.
- ▶ The 2024 flood inundation raster was used to assess:
 - ▶ Village-wise affected settlement areas (in hectares and percentage)
 - ▶ Frequency of inundation events
- ▶ Resulted in detailed village-level flood impact maps for 2024
- ▶ **Model Reliability (Validation Results):**
- ▶ **AUC (Area Under Curve) for flood inundation layer: 0.965**
 - ▶ Indicates excellent model performance
- ▶ **AUC for flood vulnerability zonation map: 0.789**
 - ▶ Indicates acceptable and reliable prediction accuracy
- ▶ The use of AHP with an 11×11 decision matrix ensured systematic weighting of parameters such as:
elevation, slope, rainfall, drainage density (DD), distance from river (DFR), TWI, SPI, LULC, NDVI, lithology, and roughness index

Recommendations / Applications

- ▶ The results can support local flood risk mitigation strategies
- ▶ Useful for infrastructure planning and development, especially in high-risk zones
- ▶ Helps improve land-use planning decisions to reduce vulnerability
- ▶ Assists in emergency preparedness and disaster management planning
- ▶ Provides a data-driven framework for policymakers and planners in flood-prone regions like Khagaria